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Program: Embedded Systems Intro

Week 9 & 10: Networked Sensors & ESP32 Web Server

Objective:

To integrate multiple peripherals (DS18B20 Temperature Sensor, OLED Display, RFID Scanner) with an ESP32 and host a local web server to display the aggregated sensor data over WiFi.

Introduction:

Real-world IoT devices require the integration of various sensors using different communication protocols (I2C for OLED, SPI for RFID, OneWire for DS18B20). This lab demonstrates how to manage these buses simultaneously while also handling network requests via the ESP32's built-in WiFi capabilities.

1. Theoretical Concepts

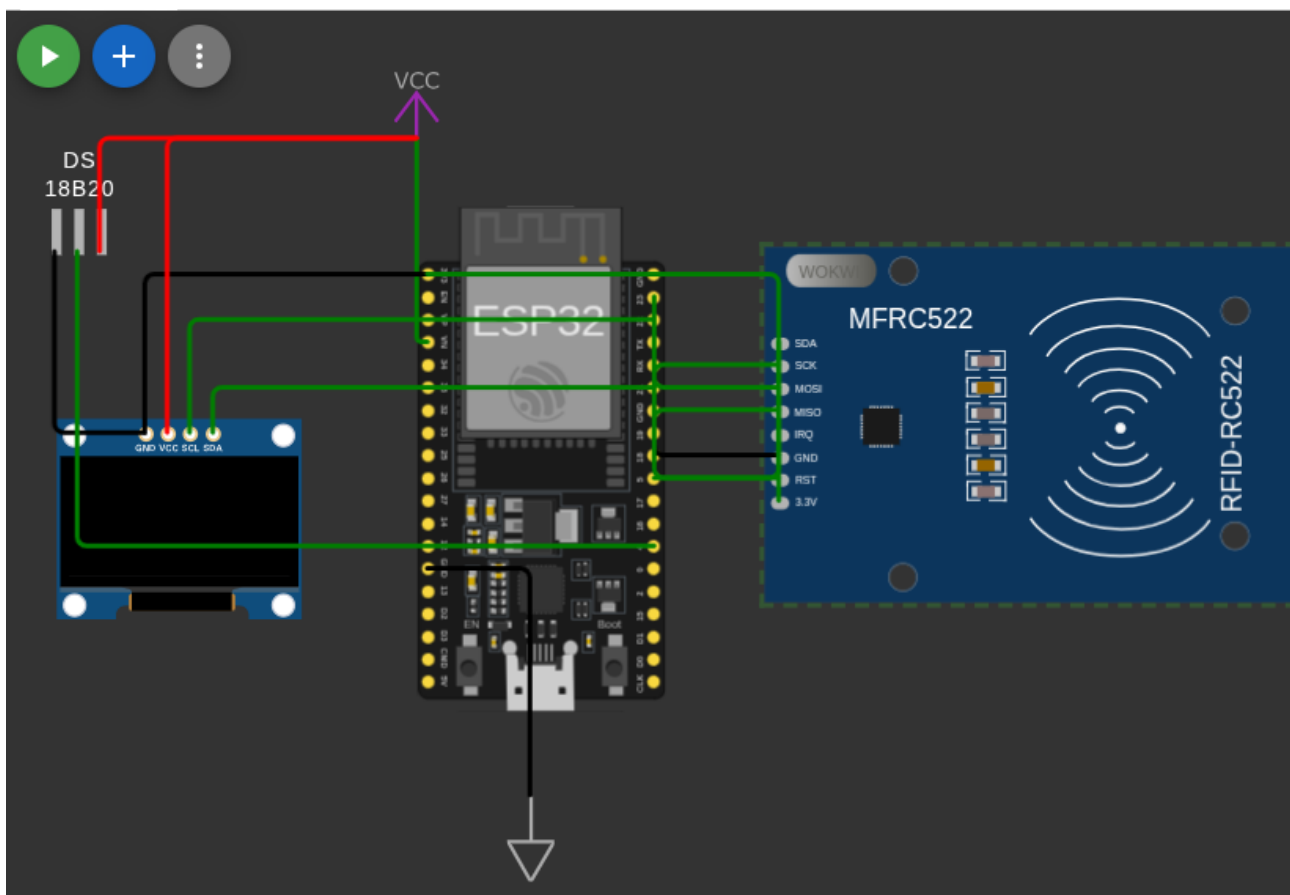
Bus Sharing & Protocols: The ESP32 is capable of managing multiple protocols simultaneously. The OLED uses the I2C bus (SDA/SCL). The DS18B20 uses the OneWire protocol which allows multiple sensors on a single data line. The MFRC522 RFID reader uses the faster SPI bus, which requires a dedicated Slave Select (SS) pin to coordinate data flow.

Asynchronous Web Serving: The ESP32 WebServer library listens for incoming HTTP GET requests on port 80. When a client connects, the ESP32 dynamically generates an HTML string containing the latest sensor readings and serves it back to the browser.

2. Source Code - ESP32 Server Integration

```
sketch.ino • diagram.json • Library Manager libraries.txt •
1 // Week 9 & 10: ESP32 IoT Server
2 // Combines a DS18B20 Temperature Sensor, an MFRC522 RFID reader, an OLED display,
3 // and a Web Server into a single integrated ESP32 application (Wokwi Compatible).
4
5 #include <Wire.h>
6 #include <SPI.h>
7 #include <Adafruit_GFX.h>
8 #include <Adafruit_SSD1306.h>
9 #include <MFRC522.h>
10 #include <WiFi.h>
11 #include <WebServer.h>
12 #include <OneWire.h>
13 #include <DallasTemperature.h>
14
15 // -----
16 // PIN DEFINITIONS (Matching standard Wokwi ESP32 wiring)
17 // -----
18
19 // DS18B20 Temperature Sensor (OneWire)
20 #define ONE_WIRE_BUS 4
21
22 // RFID Pins (SPI: SCK=18, MISO=19, MOSI=23)
23 #define SS_PIN 5
24 #define RST_PIN 22
25
26 // OLED Display (I2C: SDA=21, SCL=22)
27 #define SCREEN_WIDTH 128
28 #define SCREEN_HEIGHT 64
29
30 // -----
```

3. Hardware Integration (OLED & RFID)



4. ESP32 Local Web Server Dashboard



ESP32 IoT Sensor Dashboard

Temperature: 25.40 °C

Last RFID Scanned: 04 EA 34 52

5. Reflection

Combining I2C, SPI, OneWire, and WiFi functionality in a single loop required careful attention to blocking code. If the code spent too much time waiting for an RFID tag, reading the OneWire sensor, or drawing to the OLED, the web server would become unresponsive to incoming HTTP requests. Keeping the loop fast and non-blocking was key to a successful IoT device.